Iteration or Loops in Java

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Loops means repetition of statements depending on exit condition. In other words "repetition of execution of statement continues till the condition is true".

17 * 1 = 17
17 * 2 = 34
17 * 3 = 51
…
…

```
package Chapter4.WhileLoop;

/**
 * Write a description of class MultiplicationTable here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class MultiplicationTable
{
   public static void main(String args[]){
      //Print multiplication table of a number up to N times
      System.out.print("\f");
      int p, num = 17, start = 1;
      p = num * start;
      System.out.print("\n"+num + " * " + start + " = " + p);
      start++;
      p = num * start;
      System.out.print("\n"+num + " * " + start + " = " + p);
      start++;
      p = num * start;
      System.out.print("\n"+num + " * " + start + " = " + p);
      start++;
      p = num * start;
      System.out.print("\n"+num + " * " + start + " = " + p);
      start++;
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      System.out.print("\n"+num + " * " + start + " = " + p);
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      p = num * start;
      System.out.print("\n"+num + " * " + start + " = " + p);
      start++;
      p = num * start;
      System.out.print("\n"+num + " * " + start + " = " + p);
      start++;
      p = num * start;
      System.out.print("\n"+num + " * " + start + " = " + p);
```

these lines are repeated multiple times

```
        start++;
        p = num * start;
        System.out.print("\n"+num + " * " + start + " = " + p);
    }
}
```

Code with while loop:

```
package Chapter4.WhileLoop;
import java.util.*;

/**
 * Write a description of class MultiplicationTable here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class MultiplicationTable
{
    public static void main(String args[]){
        //Print multiplication table of a number up to N times
        System.out.print("\f");
        Scanner sc = new Scanner(System.in);
        int p, num , start = 1, endPoint;
        System.out.print("\nEnter your number: ");
        num = sc.nextInt();
        System.out.print("\nEnter end point: ");
        endPoint = sc.nextInt();
        while(start <= endPoint){
            p = num * start;
            System.out.print("\n"+num + " * " + start + " = " + p);
            start++;
        }
        System.out.print("\nValue of start after the loop: " + start);
    }
}
```

Syntax of while loop:

```
while(<condition>){
    //statements
}
```

```
public class Q1a
{
    public static void main(String args[]){
        System.out.print("\f");
        int start, no=0;
        start = 1;

        while(start<10){
            no = start*start;
            System.out.print(no + ", ");
            start++;
        }
        no = start*start;
        System.out.print(no);
    }
}
```

"\f" ← clear the terminal window

Start = 1
no = 1

output:
1,